

IBRAHEEM EL SHEIKHA

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Education

University of Toronto

Sept. 2023 – Apr. 2028

B.A.Sc. Computer Engineering, PEY Co-op

Toronto, ON

- **Top 30 ECE Student** out of ~450 students (2025–2026) · **Dean's Honours List** (2023–2026) · **J. Edgar McAllister Research Fellowship**, 1 of 2 recipients across all 1st–3rd year ECE applicants (2025)

Technical Skills

- **Languages:** C++, C, Embedded C, Python, Verilog, RISC-V Assembly, Tcl, Bash
- **Hardware & Embedded:** RTL design, logic synthesis, computer architecture, FPGA design flows, STM32, UART/SPI/I2C
- **Tools:** Quartus Prime, ModelSim, OneSpin, Git, Linux, Docker, Flask, Jira, Confluence

Experience

FPGA Software Engineering Intern

May 2026 – Present

Microchip Technology | 16-month PEY

Toronto, ON

- Develop and migrate core **C++** functionality in the **synthesis** stage of internal FPGA design flows, moving legacy code paths onto modernized implementations.
- Debug interoperability failures between third-party **EDA** libraries and internal tooling, tracing root causes through **multi-million-line Verilog netlists** spanning DSP blocks, inferred memories, and multiple clock domains.
- Build **Python** netlist post-processing utilities and automated **Bash/Tcl** test infrastructure validating end-to-end design flows across benchmark designs.

Undergraduate Researcher

May 2025 – June 2026

University of Toronto ECE | Supervised by Prof. Salma Emara

Toronto, ON

- Built the hybrid retrieval pipeline behind **PiazzaPlus**, a published **Chrome extension**: **BM25** prefilters, then cosine similarity over **OpenAI embeddings** in **ChromaDB** reranks, served via a **Flask** API.
- Designed and ran evaluation against Piazza's native search, improving **MAP** by **102.5%**, **MRR** by **143.5%**, and **P@3** by **72.4%**.
- Captioned post images with a **vision model** and added them to the index, making diagrams and screenshots searchable with text.

Projects

Varphi CPU | Verilog, ModelSim, Computer Architecture

July 2026 – Present

- Designing a configurable **finite state machine processor** that executes a Turing machine description language directly in hardware, with **no program counter and no ALU**.
- Built a **Harvard datapath** whose transition-rule ROM is addressed by state plus each tape's read symbol, parameterized over tape count with **generate blocks** so one source elaborates to single- or multi-tape.
- Wrote per-module and top-level **testbenches** plus a **Python** visualizer that animates tape head movement from simulation logs.

STM32 Perimeter Defense System | Embedded C, STM32, ESP32, Python

Mar. 2026 – Apr. 2026

- Built the sweep and ranging subsystem of a team radar turret: a stepper sweeps **360°** at 1/8 microstepping, firing an ultrasonic ping every 20 microsteps for **4.5°** resolution.
- Streamed polar contacts over **UART** at 115200 baud to a **Matplotlib** radar display, with status on an **SSD1306 OLED** over **I2C**.
- Integrated **identify-friend-or-foe** over an IR link, with an **ESP32** transmitting coded pulse frames classified after a 50 ms quiet gap.

Embedded C Reversi | Embedded C, RISC-V, DE1-SoC FPGA

Mar. 2025 – Apr. 2025

- Built a two-player Reversi game in **embedded C** on a **RISC-V soft processor**, driving VGA and a PS/2 mouse through **memory-mapped I/O**.
- Implemented **double-buffered rendering** with vertical-sync polling off the VGA status register for a tear-free **60 Hz**.
- Decoded **PS/2 mouse packets** from the data register's RVALID field and wrote the game logic: move validation, directional tile flipping, score rendering, and win detection.

Custom RISC-V CPU | Verilog, Quartus Prime, ModelSim

Oct. 2024 – Dec. 2024

- Implemented a **RISC-V-inspired CPU from scratch**: register file, **ALU**, sign-extension unit, controller FSM, datapath muxing, and a memory controller over a 512x32 RAM.
- Designed the **VGA display interface** and the memory arbitration letting processor and display adapter share one RAM; verified against directed testbenches in **ModelSim**.
- Built the ALU to cover **addi, add, sub, xor, and, or, sw, lw, beq, bne**; synthesized in **Quartus Prime Lite**.

Publications

Work-in-Progress: PiazzaPlus—Enhancing Piazza Search with Semantic Matching

June 2026

- **First author** with Prof. Salma Emara.
- Presented at the **ASEE Annual Conference & Exposition** (4,000+ attendees) — one of few undergraduate presenters.